3D Mapping Based on LiDAR Signal Processing

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Obsah

Introduction

Commercially available 3D LiDAR solutions:

Unitree 4D L1

Price: 250 €

Solution: use a cheaper 2D LiDAR and design a stepper-motor-driven platform for it

STL-19P

Price: 80 €

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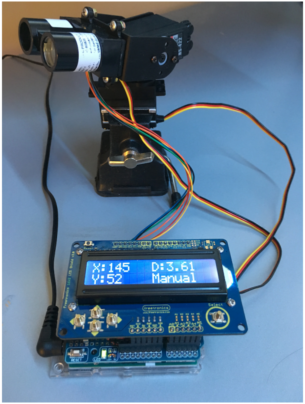
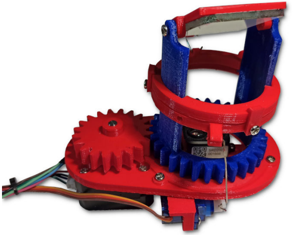
Existing Amateur Solutions for 3D Scanning



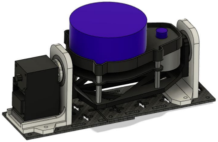
Single-Point LiDAR + Platform

- \+ Low price
- − **Slow scanning**



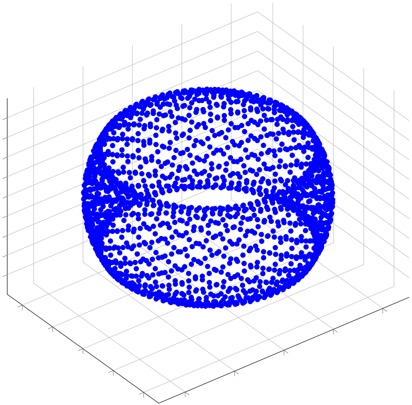
Single-Point LiDAR + Moving Mirror

- \+ Low price
- \+ LiDAR is stationary (easier to correct the distance measurement)
- − High quality mirror is needed
- − **Vibrations of the structure**

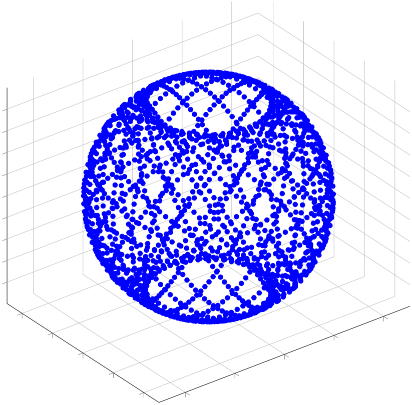


2D LiDAR + Tilting Platform

- \+ High speed
- \+ High resolution
- − **Uneven point density**



(a) $\theta = 30^\circ$

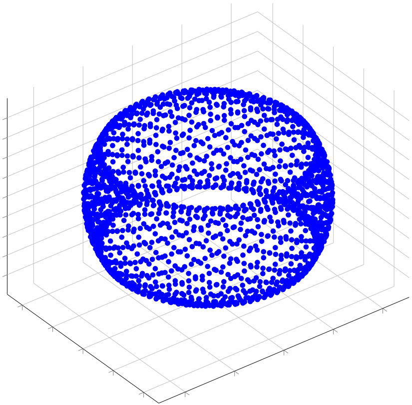


(b) $\theta = 60^\circ$

Larger tilt angle θ – more coverage

Problem: Increasing spacing between consecutive LiDAR sweeps as θ increases

Point Cloud Visualisation at Different Tilt Angles



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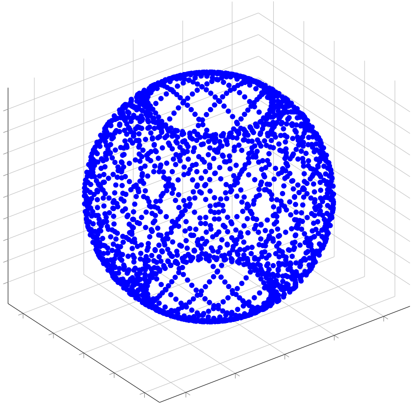

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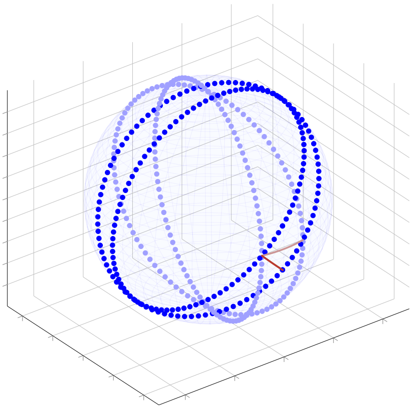
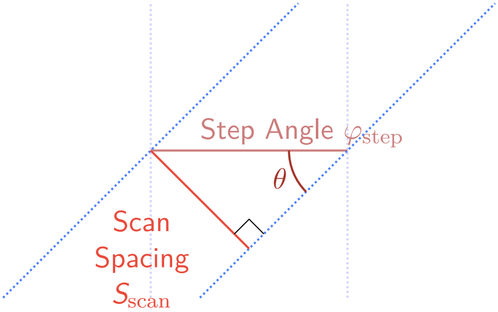
Dependence of Scan Spacing S_{scan} on Tilt Angle θ



(a) 3D visualisation



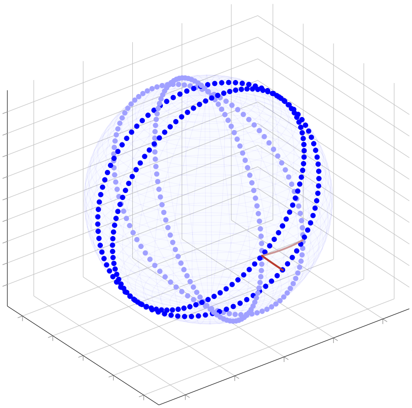
(b) Simplified model showing the dependence

Optimal tilt angle θ: 23.4°

Problem: Limited scene coverage

Solution: Design a gearbox to reduce the effective step angle

Dependence of Scan Spacing S_{scan} on Tilt Angle θ



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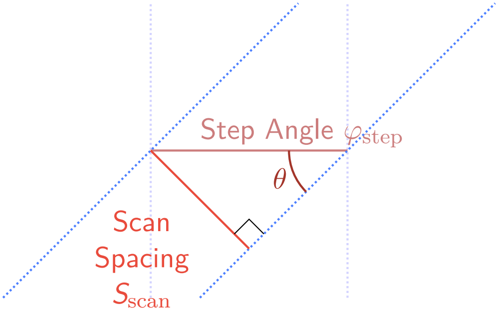

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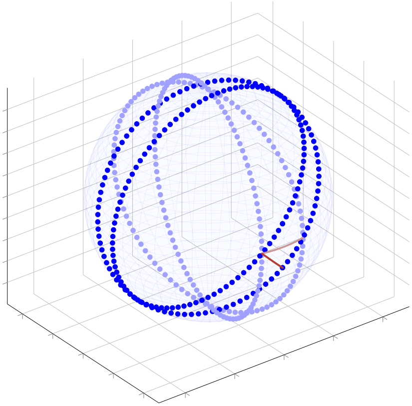

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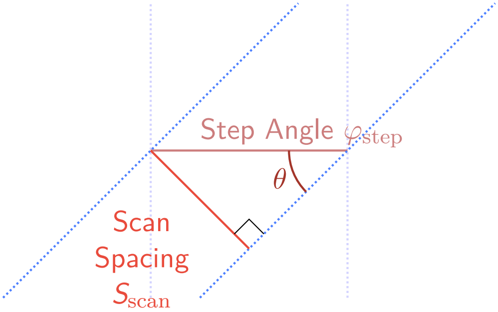

(b) Simplified model showing the dependence

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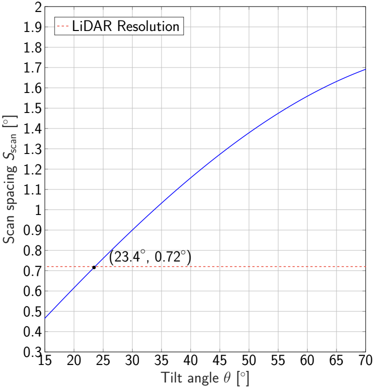
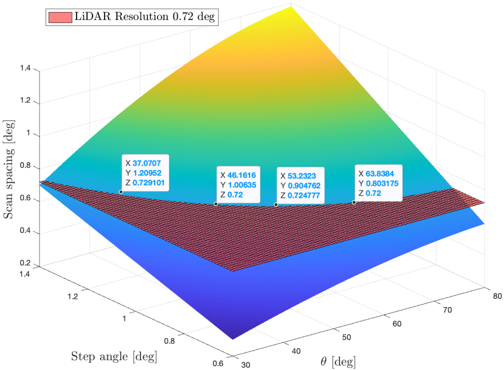
Effect of Reduction Ratio R on Scan Spacing S_{scan}

$$S_{\text{scan}} = \varphi_{\text{step}} \cdot \sin(\theta)$$

$$\varphi_{\text{eff}} = \frac{\varphi_{\text{step}}}{R}, S_{\text{scan}} = \varphi_{\text{eff}} \cdot \sin(\theta)$$

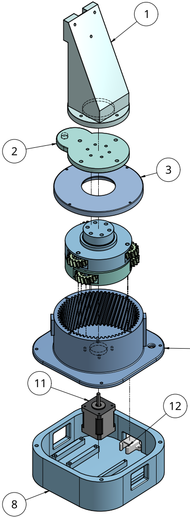


Without a gearbox

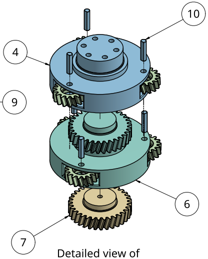


With a gearbox

Platform Showcase



Item	Quantity	Description
1	1	Platform
2	1	Platform Mount
3	1	Ring Gear Lid
4	1	Upper Carrier
5	6	CCW Gear
6	1	Lower Carrier
7	1	Motor Shaft Gear
8	1	Bottom
9	1	Ring Gear
10	6	Rod
11	1	NEMA 17
12	1	DC Barrel Jack Mount



Detailed view of
the insides of gearbox

Rigid structure

High reduction ratio $R = 18$ ($\varphi_{\text{eff}} = 0.1^\circ$) $\rightarrow$ denser point cloud & ability to choose different tilt angles while maintaining the uniform point cloud density

Total price of the platform (with LiDAR and electronics): 164 €

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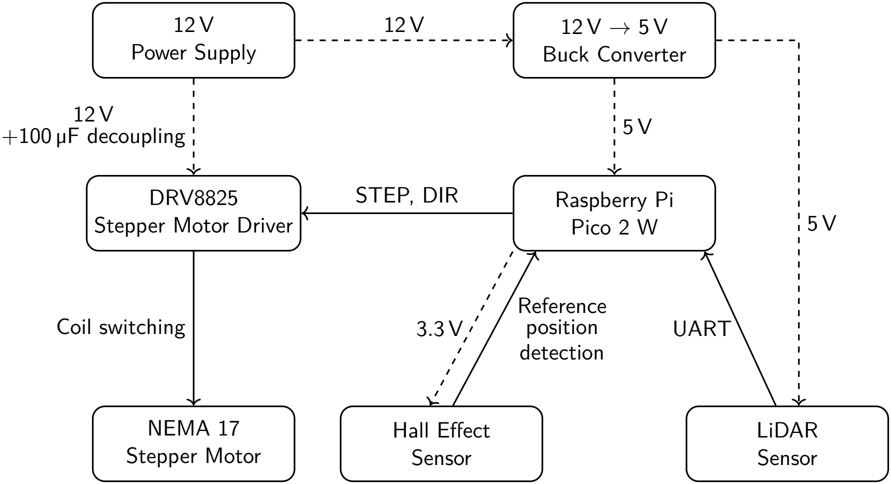
Electronic Design



Software Architecture

LiDAR $\xrightarrow{\text{UART}}$ Pico 2 W $\xleftrightarrow{\text{Wi-Fi}}$ PC $\longrightarrow$ Open3D

Pico 2 W

- Homing and motor control
- UART acquisition
- Wi-Fi transmission

PC Receiver Thread

- Packet reception
- 3D reconstruction

PC Main Thread

- Open3D visualisation
- Point cloud export

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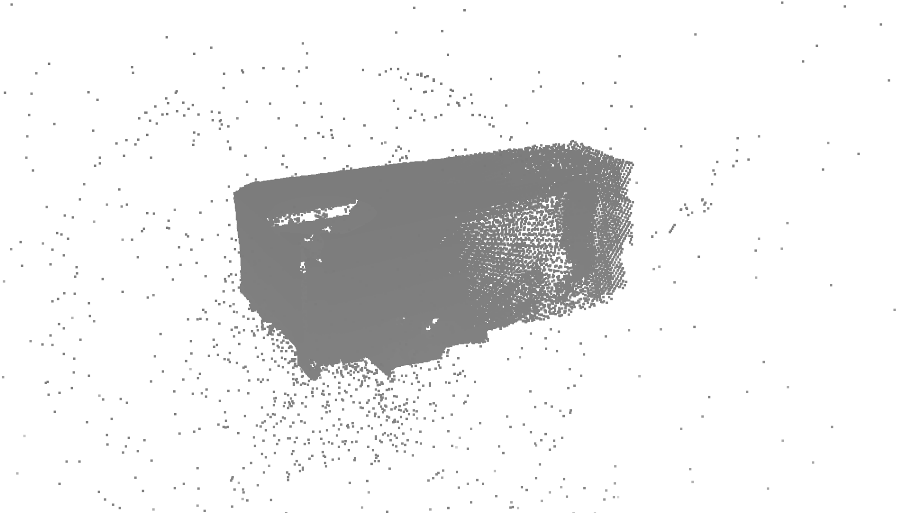
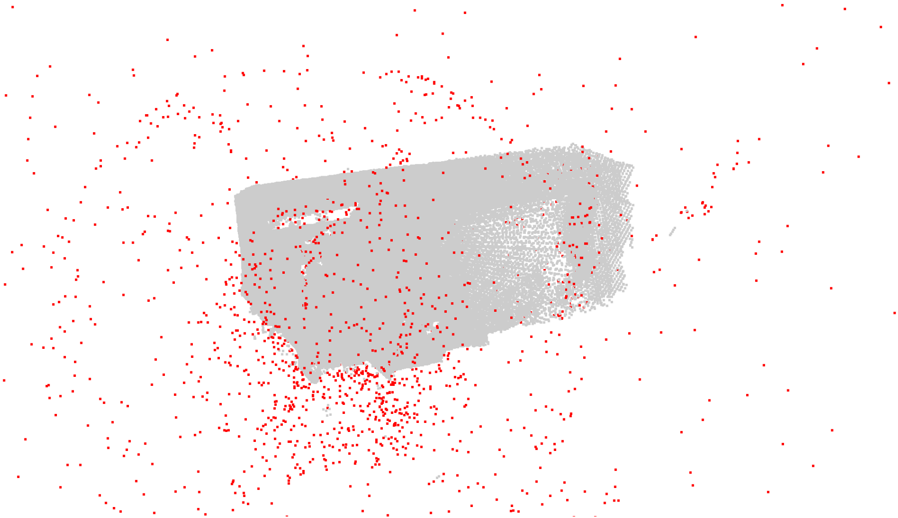
Room Scan

Tested with $\theta = 64^\circ$ and 0.8° step angle between LiDAR sweeps.



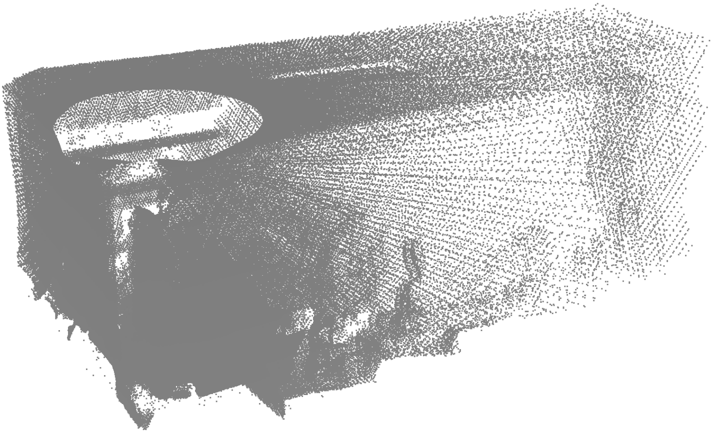
Raw point cloud

Room Scan



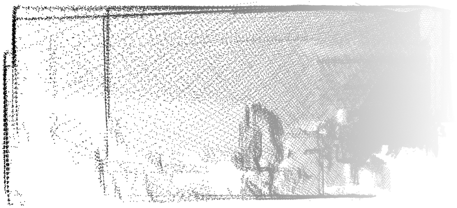
Detected outliers

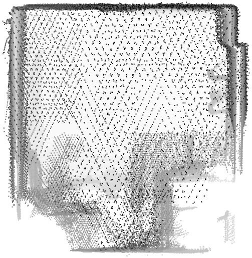
Room Scan

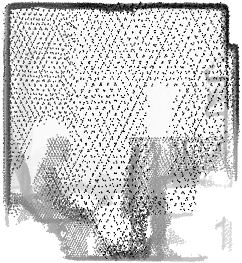


Filtered point cloud

Transformation Corrections

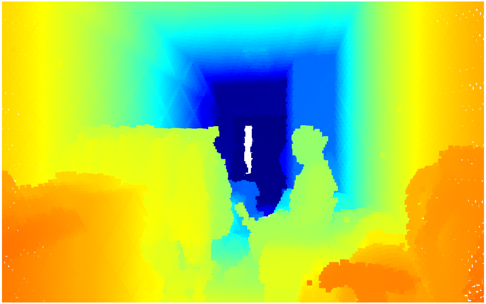






Comparison with a Photograph

Photo of a room



Resulting point cloud

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Conclusion

A platform enabling 3D reconstruction with 2D LiDAR was built. It features a gearbox with a high reduction ratio, allowing for dense point clouds even at large tilt angles.

Code for data acquisition, processing and 3D reconstruction visualisation was written.

The prototype was tested by scanning a room, transformation corrections were applied.

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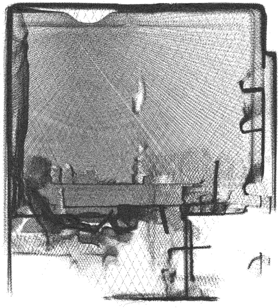
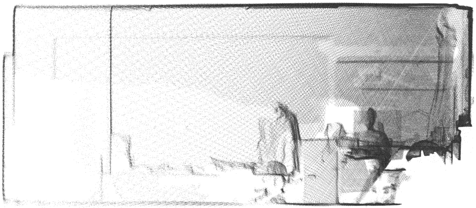
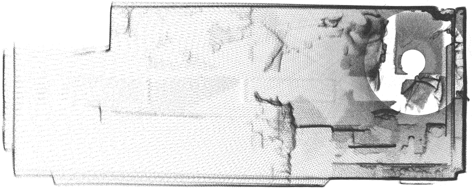
Odpovědi na otázky

Q1:Jaká je opakovatelnost měření, zejména s ohledem na zarovnání s využitím Hallovy sondy při inicializaci měření?

A1.1: Měření bylo opakováno 10krát, point cloudy byly překryty.

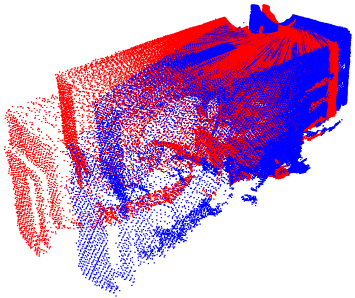


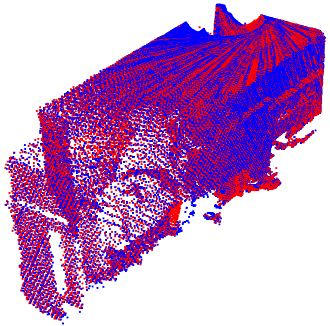




Odpovědi na otázky

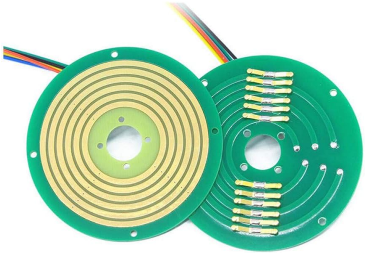
A1.2: Open3D nabízí spoustu funkcí pro zarovnávání a porovnávání point cloudů. Point-to-Point Iterative Closest Point (ICP) algoritmus byl použit pro demonstrační účely.





Odpovědi na otázky

Q2: Jaké mechanické úpravy by byly potřeba pro umožnění nepřetržité rotace senzoru? Jakým způsobem by bylo možné kontinuálně scanovat bez nutnosti inicializace orientace před každým spuštěním?



Cena: >100 $. Odkaz na Amazon

A2: V LiDARech se používají tzv. slip rings pro umožnění přenosu dat a napájení při otáčení o 360°. Neintegrovaná řešení jsou avšak poměrně drahá.

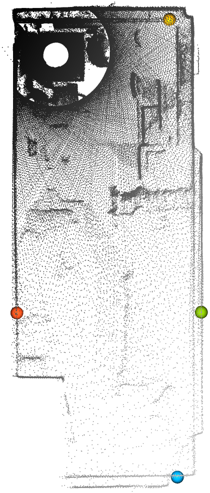
Odpovědi na otázky

Q3: Jaká je časová náročnost provedení celého 3D scanu?

A3: $450\,\text{kroků} \cdot 0.105\,\text{s/krok} + 10\,\text{s(homing)} = \approx 60\,\text{s}$ Reálná doba je však o cca 10 s delší kvůli přenosu dat přes Wi-Fi. V ojedinělých případech může dojít ke ztrátě spojení nebo komunikačním prodlevám, což může dále prodloužit dobu skenování.

Odpovědi na otázky

Q4: Odpovídají rozměry nascanovaného pointcloudu reálným rozměrům místnosti? Jaké je relativní zastoupení outlierů v získaném pointcloudu?



Rozměry pokoje: $5.87 \pm 0.05\text{m} \times 2.35 \pm 0.05\text{m}$ (vzato z wiki Silicon Hill)

P0, P1, P2, P3

```
|P1 - P0|: 5.9352 <- length
|P2 - P1|: 3.0441
|P3 - P2|: 2.3780 <- width
```

Rozměry nascanovaného pointcloudu jsou v rámci chyby měření

```
Unfiltered pcd: 494400  points
Outliers:       3737    points
Filtered pcd:   490663  points
Removed points percentage: 0.76%
```